Shreya Harish Yagain

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Education

RV College of Engineering **2024 – 2028**
Bachelor of Engineering in Electronics and Communication (CGPA: 7.93/10) *Bengaluru, India*

Narayana PU College **2023 – 2024**
Class XII (90%) *Bengaluru, India*

National Hill View Public School **2021 – 2022**
Class X (90%) *Bengaluru, India*

Internship Experience

Velynk | *College Outreach & Platform Testing Intern* **September 2026 – Present**

- Tested the platform's user flows end to end before launch, documenting usability issues and giving actionable feedback.
- Lead outreach to faculty, clubs, students, and prospective mentors, managing follow-ups to drive early adoption.
- Plan upcoming AMA sessions and a hackathon, creating promotional materials and coordinating with stakeholders.

RACE EL Systems | *PCB Design & Signal Integrity Intern* **August 2026 – September 2026**

- Studied signal integrity issues (reflections, ringing, overshoot, undershoot, crosstalk) and how to mitigate each.
- Learned how PCB stack-up, trace geometry, dielectrics, routing topology, and termination affect high-speed signal quality.
- Applied these concepts through hands-on analysis and simulation in Ansys HFSS.

Projects

Hardware-Based Memory Controller with Data Bus Inversion | *Verilog HDL, Vivado* | GitHub **2025**

- Designed a Verilog memory controller using Data Bus Inversion (DBI) to cut bit transitions and dynamic power.
- Simulated in Xilinx Vivado, measuring 35% fewer bit transitions and 28% lower dynamic power.

Dual-Mode Personal Portfolio Website | *React, Vite, Framer Motion, GSAP, Tailwind, CSS* | Portfolio **2026**

- Built a fully responsive dual-sided portfolio with a 3D card-flip toggle between an oscilloscope-themed engineering view and a scrapbook art view.
- Developed custom SVG electronics visualizations and scroll-triggered animations.

Wearable Hyperkalemia Monitor | *ESP32, AD8232, Python, Machine Learning, C++* **2026**

- Built a wearable ECG device (AD8232 + ESP32) streaming live signals over Wi-Fi to a Python classifier.
- Flags hyperkalemia risk from ECG features on a live dashboard; extending it to detect hypokalemia.

Technical Skills

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL
Databases: MySQL
Web Development: HTML, CSS, React.js, Node.js
Hardware & Embedded: Arduino, ESP32, Verilog HDL, Vivado, Ansys HFSS, Digital System Design
Tools & Platforms: Git, GitHub, Visual Studio Code, LaTeX, Figma

Certifications

NPTEL: Database Management Systems, Data Science for Engineers, Data Structures & Algorithms
Samsung Innovation Campus: Data Structures & Algorithms using Python

Leadership / Activities

Sattva Art Club | *Social Media Head, RVCE* **2025 – Present**

- Led creative strategy and content design to enhance digital presence and student engagement.

Volunteering | *Govt. School, Amrutha Shishu Nivasa, People for Animals* **Bengaluru**

- Volunteered 30 hours teaching at a government school and 40 hours at an orphanage; supported care-taking, record-keeping, rescues, and adoption at an animal rescue center.